Unit 3

3. Machine Learning (ML) – Overview

Module 3.2

3.2 ML Workflow

Tutorial 3.2

3.2 ML Workflow

AI-3.2.1 K2 Summarize the workflow used to create an ML system.

What Is an ML Workflow?

- Machine Learning workflow is a structured sequence of steps used to build, train, validate, and deploy a machine learning model.
- Key Idea
  - Just like software development has SDLC, ML systems follow a repeatable workflow which is known as ML workflow
  - Each step affects the model’s performance, reliability, and testability
  - Testing must be integrated throughout the workflow and not just at the end

ML Workflow

Image Source - ISTQB CT AI Syllabus

ML Workflow (Setting the Direction)

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1. **Understand the Objectives**
   - Clearly define the business problem the ML system will solve
   - Identify success criteria (accuracy, fairness, speed, etc.)
   - Understand constraints (budget, regulations, timelines)
   - Align objectives with stakeholder expectations
2. **Select a Framework**
   - Choose an ML development environment or library (e.g., TensorFlow, PyTorch)
   - Ensure it supports the type of ML task (classification, regression, clustering)
   - Check compatibility with hardware and deployment needs
   - Consider community support and available tools

ML Workflow (Designing the Foundation)

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3. **Select & Build the Algorithm**
   - Pick an algorithm suited to the problem type
   - Configure initial parameters (hyperparameters)
   - Build a basic model structure ready for training
   - Ensure it aligns with data size and complexity
4. **Prepare & Test Data**
   - Clean data (remove errors, handle missing values)
   - Split into training, validation, and test sets
   - Ensure data is representative of real-world scenarios
   - Test data quality for bias and completeness

ML Workflow (Building & Evaluating)

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5. Train the Model

- Feed training data into the model
- Adjust internal parameters to learn patterns
- Repeat until performance is acceptable on training data
- Monitor for overfitting during training

6. Evaluate the Model

- Test on validation data to check generalization
- Use metrics like accuracy, precision
- Identify strengths and weaknesses of the model
- Compare results against baseline performance

ML Workflow (Optimizing & Testing)

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7. Tune the Model

- Adjust hyperparameters to improve performance
- Try different algorithms or architectures if needed
- Use cross-validation to confirm improvements
- Avoid overfitting while optimizing

8. Test the Model

- Use the test dataset for final performance check
- Validate against edge cases and rare scenarios
- Check for bias, fairness, and robustness
- Ensure compliance with regulatory requirements

ML Workflow (Go Live)

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9. Deploy the Model

- Integrate into the target application or system
- Ensure it works in the real-world environment
- Provide documentation for users and maintainers
- Set up monitoring tools for live performance

10. Use the Model

- Allow end-users or systems to make predictions
- Ensure input data format matches training data
- Monitor for unexpected outputs during use
- Gather user feedback for improvements

ML Workflow (Continuous Improvement)

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11. Monitor and Tune the Model

- Track performance over time in production
- Detect concept drift or changes in data patterns
- Retrain or fine-tune to maintain accuracy
- Document all changes and updates for traceability

Testing in the ML Workflow

- Validate Data
  - Check for bias, completeness, and relevance
- Evaluate Model
  - Use metrics like accuracy, precision, recall, MSE
- Behavior Testing
  - Simulate edge cases, adversarial inputs, and concept drift
- Explainability Checks
  - Ensure decisions can be interpreted and justified
- Post-deployment Monitoring
  - Detect performance degradation and retraining needs

THANK YOU!!!